



**IILM UNIVERSITY**  
GREATER NOIDA,  
UTTAR PRADESH

#### UNIVERSITY INTERNSHIP REPORT

# Web Development Internship

Event Management & URL Shortener

#### INTERNSHIP DURATION

1st August 2026 to 30th August 2026

#### STUDENT CREDENTIALS

**Tushar Jha**

Enrollment / Roll No.  
25scs1003004469

Section & Program  
2CSE33 | B.Tech (CSE)

#### HOST ORGANIZATION

**CodeAlpha**

Virtual Backend Development Internship

# Candidate & Program Overview

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- **Student Name:** TUSHAR JHA
- **Roll Number:** 25scs1003004469
- **Section:** 2CSE33
- **Degree:** B.Tech (Computer Science & Engineering)
- **Batch:** 2025-29
- **University:** IILM University, Greater Noida, U.P.
- **Internship Domain:** Backend Development



# Problem Statement

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## Event Coordination

Organizations face persistent challenges in coordinating multiple variables associated with event hosting, resulting in fragmented communication and data silos.



## Link Management

The modern web requires efficient link sharing; long URLs are cumbersome, prone to truncation, and difficult to track for analytics.



## Robust Utilities

There is a strong, persistent need for accessible, reliable digital tools to solve both event management and URL optimization issues efficiently.

# Core Project Objectives

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**Event Management Integration:** To design, develop, and integrate a comprehensive Event Management web application tailored for ease of user navigation and streamlined event coordination.



**URL Shortener Utility:** To build a reliable URL Shortener that generates unique aliases and efficiently maps them back to original lengthy URLs for improved sharing.



**API Construction:** To construct scalable RESTful APIs that serve both systems, handle client requests securely, and manage database operations efficiently.



**Full Lifecycle Scope:** Encompass the full software development life cycle for both applications, including user management, link creation, and redirection handling.

# Agile Development Methodology

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## Sprint 2 - Backend

Building scalable RESTful API endpoints and robust core server logic operations using Node.js.

## Sprint 4 - Deployment

Culminating in thorough testing, rigorous debugging phases, and final application deployment.

## Sprint 1 - Planning

Initial focus on requirements gathering, task scoping, and designing logical database schemas.

## Sprint 3 - Integration

Integrating the React.js frontend interfaces with backend services for seamless user interaction.

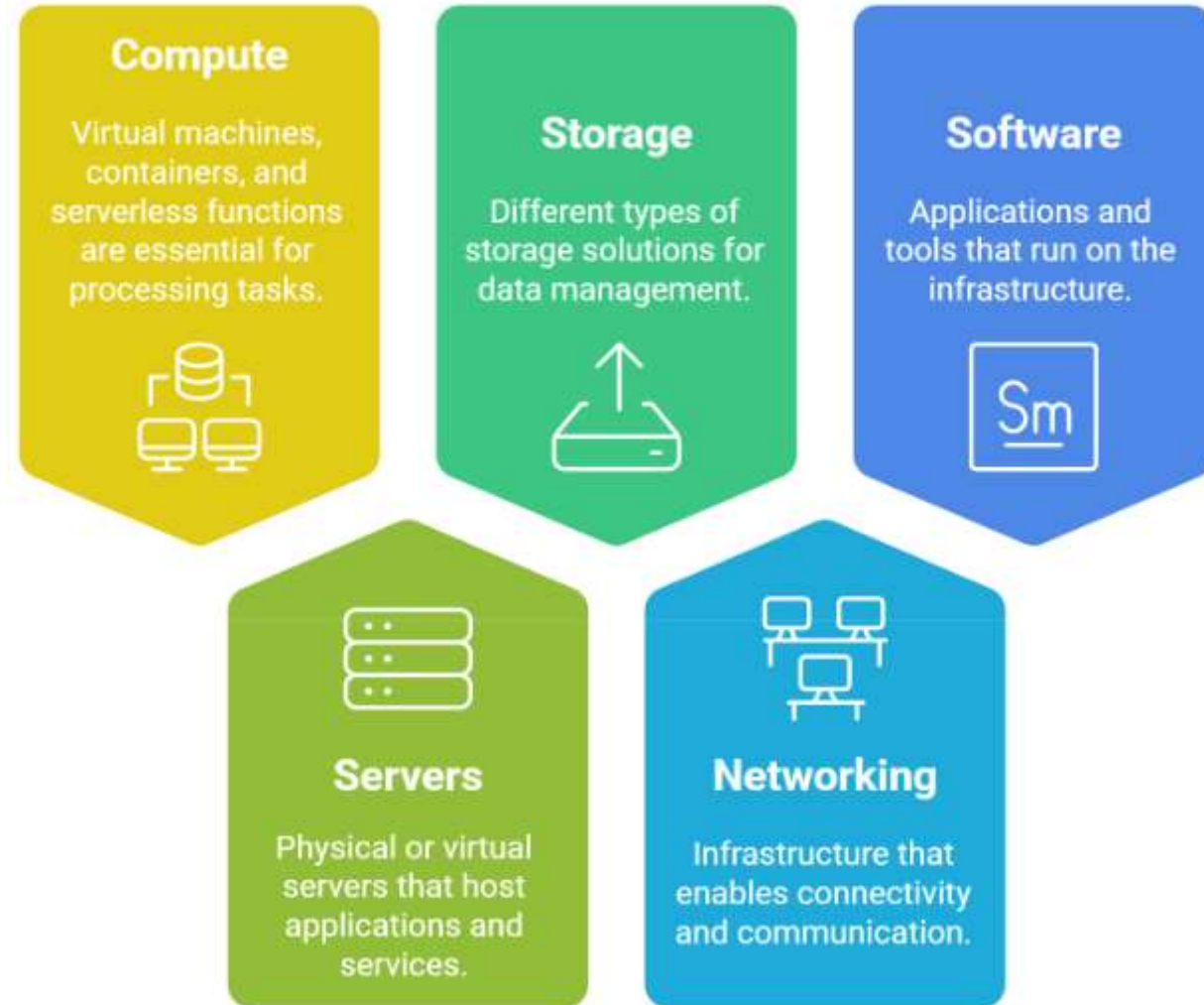
# System Architecture

## Client-Server Model

The deployed applications employ a strict Client-Server architecture. The React.js user interface communicates asynchronously with the Node.js backend server via HTTP protocols and RESTful endpoints.

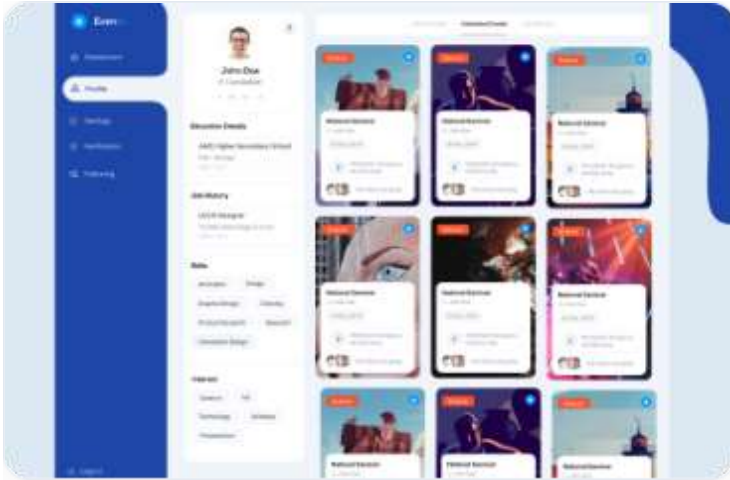
The backend logic securely interacts with the MongoDB/PostgreSQL database layer to persist state, such as saving complex event details or securely mapping a short token to its corresponding lengthy URL destination.

## Cloud Infrastructure Components



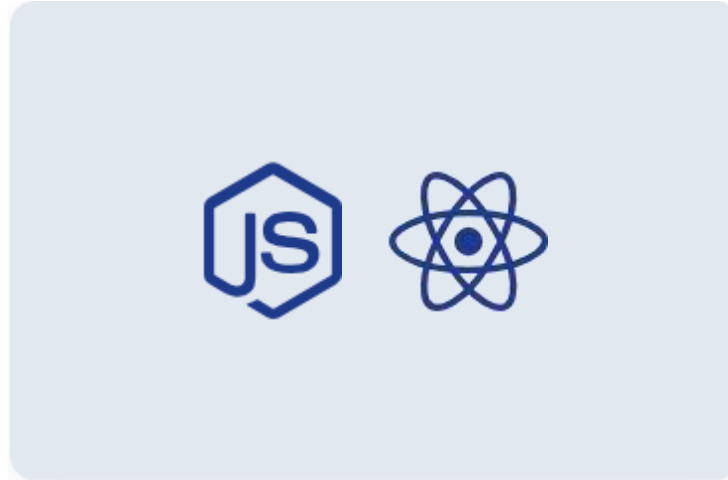
# Technologies & Outcomes

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## Deployed Codebases

Fully functional and hosted codebases available for review, reflecting standard industry practices.



## Tech Stack Mastery

Demonstrated proficiency in JavaScript, HTML/CSS, React.js, Node.js, MongoDB/PostgreSQL, and Git.



## Practical Utility

Users can now seamlessly create coordinated events and manage web links efficiently.

# CERTIFICATE OF COMPLETION

This Certificate Is Proudly Presented To

**Tushar Jha**

This Certificate recognizes the successful completion of the CodeAlpha Virtual Internship Program in Backend Development, demonstrating dedication and sincerity throughout the program.

We appreciate the valuable efforts and wish continued success in all future endeavors.

**1st August 2026 to 30th August 2026**



CEO & Co-Founder

**1st September 2026**

Date of Issue



SCAN TO VERIFY THIS CERTIFICATE  
Student ID: CA/DFI/229260

# Thank You!

End of Presentation

 [github.com/tushar23456-eng](https://github.com/tushar23456-eng)